

Lecture 3

Linear representations and complete reducibility

Objectives. *Symmetry enters physics through linear spaces: in classical physics small oscillations add, and in quantum physics states superpose. This lecture sets up the central notion of the course — a linear representation of a group — together with the construction by which symmetries act on functions, and proves the two structural theorems that make finite group representations tractable: every representation can be made unitary by averaging, and every representation decomposes into irreducible pieces (Maschke’s theorem). The permutation representation of S_3 is decomposed in full as a model computation.*

3.1 Representations: definition and first examples

Lectures 1 and 2 studied groups in the abstract. Physics, however, meets a group through its *action on the states of a system*, and the state spaces of physics — quantum Hilbert spaces, spaces of displacements of atoms, spaces of fields — are vector spaces. The operations of the group must respect that linear structure: a symmetry operation applied to a superposition is the superposition of the transformed states. The mathematical object capturing this is the following.

Definition 3.1 (Representation). A (linear) representation of a group G on a finite-dimensional complex vector space V is a homomorphism

$$\rho: G \longrightarrow \mathrm{GL}(V),$$

where $\mathrm{GL}(V)$ is the group of invertible linear operators on V . Thus to each $g \in G$ is assigned an invertible operator $\rho(g)$, with

$$\rho(gh) = \rho(g) \rho(h) \quad \text{for all } g, h \in G,$$

whence automatically $\rho(e) = \mathrm{id}_V$ and $\rho(g^{-1}) = \rho(g)^{-1}$. The *dimension* (or degree) of the representation is $\dim V$. We speak of “the representation (ρ, V) ”, or simply of V when ρ is understood.

Unless stated otherwise our vector spaces are complex and finite-dimensional; the real case will be flagged when it matters (and it will matter — see Example 3.12).

Notation: operators versus matrices. Throughout these notes we keep two parallel notations, and the distinction is deliberate. The symbol $\rho(g)$ always denotes the abstract

linear operator on V , defined without reference to any basis. Once a basis $(e_1, \dots, e_n)$ of V is chosen, the corresponding matrices are denoted $D(g)$ and defined by

$$\rho(g) e_j = \sum_{i=1}^n D_{ij}(g) e_i.$$

The j -th column of $D(g)$ lists the components of the image of the j -th basis vector. The homomorphism property becomes matrix multiplication, $D(gh) = D(g)D(h)$, and a map $D: G \rightarrow \text{GL}(n, \mathbb{C})$ with this property is called a *matrix representation*. Irreducible representations, once we have them, will carry labels $\rho^{(i)}$, $D^{(i)}$. Keeping operators and matrices typographically apart costs little and prevents a great deal of confusion when several bases are in play at once — as they constantly are from Section 3.6 onwards.

Example 3.2 (One-dimensional representations). On a one-dimensional space, an invertible operator is multiplication by a nonzero number: $\text{GL}(1, \mathbb{C}) = \mathbb{C} \setminus \{0\}$. A one-dimensional representation is therefore nothing but a homomorphism $G \rightarrow \mathbb{C} \setminus \{0\}$, and we have already met several:

- (i) the *trivial representation* $\rho(g) = 1$ for all g , defined for every group, on which symmetry acts invisibly;
- (ii) the *sign representation* $\text{sgn}: S_n \rightarrow \{\pm 1\}$ of Lecture 1;
- (iii) the *orientation representation* $\nu: D_n \rightarrow \{\pm 1\}$ of Lecture 2, with $\nu = +1$ on rotations and -1 on reflections.

Example 3.3 (All the χ_k of a cyclic group). Let $C_n = \langle r \rangle$ and $\omega = e^{2\pi i/n}$. For each $k \in \{0, 1, \dots, n-1\}$,

$$\chi_k(r^m) = \omega^{km} = e^{2\pi i km/n}$$

is a one-dimensional representation: $\chi_k(r^a)\chi_k(r^b) = \omega^{k(a+b)} = \chi_k(r^{a+b})$, and the value is well defined because $\omega^{kn} = 1$. These n representations are pairwise distinct ($\chi_k(r) = \omega^k$ differ), and in Lecture 4 we shall see that they are *all* the irreducible representations of C_n — the finite Fourier modes, as we will recognise when characters appear.

Example 3.4 (The defining representation of D_n). Realising D_n by orthogonal transformations of the plane, as in Lecture 1, is a two-dimensional real representation, called the *defining* (or *vector*) representation. In the basis with vertex 1 on the y -axis (Sheet 1 instead puts the mirror of s on the x -axis, $s = \text{diag}(1, -1)$; both choices satisfy the dihedral relations and are conjugate by a rotation through $\pi/2$), the generators of D_3 have matrices

$$D(r) = \begin{pmatrix} \cos \frac{2\pi}{3} & -\sin \frac{2\pi}{3} \\ \sin \frac{2\pi}{3} & \cos \frac{2\pi}{3} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}, \quad D(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Multiplying the matrices out, we check that

$$D(r)^3 = D(s)^2 = \mathbb{1}, \quad D(s)D(r)D(s)^{-1} = D(r)^{-1},$$

i.e. the dihedral relations hold. That is enough to define D on the whole group. Lecture 1 listed the elements of D_n as $r^a s^\varepsilon$ with $a \in \{0, 1, \dots, n-1\}$ and $\varepsilon \in \{0, 1\}$, and each element appears in that list exactly once. We call this way of writing an element its *normal form*. Set

$$D(r^a s^\varepsilon) = D(r)^a D(s)^\varepsilon.$$

Because the normal form is unique, this formula assigns exactly one matrix to each element, so D is well defined.

The same element can also be written in other ways. In D_3 , for instance, $sr = r^2s$. The formula reads only normal forms, so it says nothing directly about the product $D(s)D(r)$. The relations show that this product is the right matrix:

$$D(s)D(r) = D(r)^{-1}D(s) = D(r)^2D(s) = D(r^2s).$$

The same happens for every product. D is multiplicative, because the same three relations that give Lecture 1's product rule $(r^a s^\epsilon)(r^b s^\delta) = r^{a+(-1)^\epsilon b} s^{\epsilon+\delta}$ hold for the matrices.

Groups often arrive acting on a set before they act on a vector space; the following definition names what we have in fact been doing since Lecture 1.

Definition 3.5 (Group action). An action of a group G on a set X is a map

$$G \times X \rightarrow X, \quad (g, x) \mapsto g \cdot x,$$

such that $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G, x \in X$.

Examples of actions: symmetry operations acting on the points of space; S_n acting on $\{1, \dots, n\}$; and G acting on itself by conjugation (Lecture 2).

Example 3.6 (Permutation representations). Let G act on a finite set X . Take a vector space with one basis vector e_x for each point $x \in X$, namely $V = \mathbb{C}^X$, and let G permute the basis the way it permutes the points:

$$\rho(g) e_x = e_{g \cdot x}.$$

Then $\rho(g)\rho(h)e_x = e_{g \cdot (h \cdot x)} = e_{(gh) \cdot x} = \rho(gh)e_x$: a representation of dimension $|X|$, the *permutation representation* of the action. Its matrices are *permutation matrices* — a single 1 in each row and column. For S_3 acting on $\{1, 2, 3\}$, the generators $(1\ 2)$ and $(1\ 2\ 3)$ act on the basis (e_1, e_2, e_3) by

$$D((1\ 2)) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D((1\ 2\ 3)) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

the columns recording $e_1 \mapsto e_2, e_2 \mapsto e_3, e_3 \mapsto e_1$ in the second case. This representation is decomposed in full in the tutorial. Taking $X = G$ with the action of G on itself by left multiplication yields the *regular representation*, of dimension $|G|$, whose properties are the subject of Lecture 4.

Remark 3.7 (Faithfulness). A representation is *faithful* if ρ is injective ($\ker \rho = \{e\}$), i.e. if the matrices remember the whole group. The defining representation of D_n and all permutation representations of group actions in which only e fixes every point (actions that *separate points*) are faithful; the trivial representation is maximally unfaithful. There is no requirement of faithfulness in the theory: a representation may legitimately see only a quotient $G/\ker \rho$ — recall the orientation representation, which sees only orientation.

3.2 How symmetries act on functions

We now come to the single most important construction of the course. The states of physical systems are very often *functions* on the space the symmetry group transforms: wavefunctions

on configuration space, displacement fields on a molecule, temperature distributions on a body. The group's action on points then induces an action on the functions — and it does so by a formula with one subtle feature: an inverse.

Theorem 3.8 (The function-space representation). *Let the group G act on a set X , and let $\mathcal{F}(X)$ denote the vector space of complex-valued functions on X . For $g \in G$ and $\psi \in \mathcal{F}(X)$ define*

$$(\rho(g)\psi)(x) := \psi(g^{-1} \cdot x).$$

Then each $\rho(g)$ is an invertible linear operator and $\rho: G \rightarrow \text{GL}(\mathcal{F}(X))$ is a representation. The same formula defines a representation on any subspace of $\mathcal{F}(X)$ stable under all the $\rho(g)$ (for infinite X one works with such subspaces — polynomials, smooth functions, square-integrable functions — as the context requires).

Proof. Linearity in ψ is clear, and $\rho(e) = \text{id}$. The homomorphism property is the following computation:

$$(\rho(g)\rho(h)\psi)(x) = (\rho(h)\psi)(g^{-1} \cdot x) = \psi(h^{-1} \cdot (g^{-1} \cdot x)) = \psi((gh)^{-1} \cdot x) = (\rho(gh)\psi)(x),$$

using $h^{-1}g^{-1} = (gh)^{-1}$ from Lecture 1. Invertibility follows: $\rho(g)\rho(g^{-1}) = \rho(e) = \text{id}$. ■

Why the inverse? First the conceptual answer: $\rho(g)\psi$ is the function ψ carried along by g . Evaluating the definition at a transformed point,

$$(\rho(g)\psi)(g \cdot x) = \psi(x).$$

Read it like this: the transformed function $\rho(g)\psi$ takes, at the transported point $g \cdot x$, the value that the original function ψ took at the original point x .

For example, let X be finite, and let δ_x be the function equal to 1 at the point x and to 0 elsewhere. Then $(\rho(g)\delta_x)(y) = \delta_x(g^{-1} \cdot y)$ equals 1 exactly when $g^{-1} \cdot y = x$, that is, when $y = g \cdot x$. Hence

$$\rho(g)\delta_x = \delta_{g \cdot x}.$$

The spike that sat at x now sits at $g \cdot x$: it moves together with the point. The functions δ_x play the role of the basis vectors e_x of Example 3.6.

Second, the mechanical answer: the inverse is forced by the order of composition. Had we defined $(\sigma(g)\psi)(x) = \psi(g \cdot x)$, the same computation would give $\sigma(g)\sigma(h) = \sigma(hg)$ — the products in the wrong order, an *anti*-homomorphism. The inverse undoes that reversal.

The action of permutations on polynomials, mentioned at the end of Lecture 1, is the same mechanism written in coordinate slots. Take the left action

$$\sigma \cdot (x_1, \dots, x_n) = (x_{\sigma^{-1}(1)}, \dots, x_{\sigma^{-1}(n)}).$$

It is an action, because the i -th entry of $\sigma \cdot (\tau \cdot x)$ is $x_{\tau^{-1}\sigma^{-1}(i)} = x_{(\sigma\tau)^{-1}(i)}$. Its inverse sends x to $(x_{\sigma(1)}, \dots, x_{\sigma(n)})$, so

$$(P_\sigma f)(x_1, \dots, x_n) = f(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

In words, P_σ replaces each variable x_i by $x_{\sigma(i)}$. By Theorem 3.8 this is a representation, and the composition can also be checked directly:

$$\begin{aligned} (P_\sigma(P_\tau f))(x_1, \dots, x_n) &= (P_\tau f)(x_{\sigma(1)}, \dots, x_{\sigma(n)}) \\ &= f(x_{\sigma(\tau(1))}, \dots, x_{\sigma(\tau(n))}) = (P_{\sigma\tau} f)(x_1, \dots, x_n). \end{aligned}$$

The tempting alternative replaces x_i by $x_{\sigma^{-1}(i)}$. Call it Q_σ . The same computation gives $Q_\sigma Q_\tau = Q_{\tau\sigma}$: the product comes out in the wrong order, exactly as for the rule without the inverse above.

Example 3.9 (Coordinate functions; the defining representation recovered). Let $G \subseteq O(2)$ act on the plane, and consider the two-dimensional space of *linear* functions spanned by the coordinate functions x_1, x_2 (so $x_i(\vec{r}) = r_i$). This space is stable:

$$(\rho(g)x_i)(\vec{r}) = x_i(g^{-1}\vec{r}) = (g^{-1})_{ij}r_j = (g^T)_{ij}r_j = g_{ji}x_j(\vec{r}),$$

summation over j understood and orthogonality $g^{-1} = g^T$ used. Hence $\rho(g)x_i = \sum_j g_{ji}x_j$: comparing with the convention $\rho(g)e_j = \sum_i D_{ij}(g)e_i$, the matrix of $\rho(g)$ in the basis (x_1, x_2) is exactly g itself. For $G = D_3$ the function-space representation on linear functions *reproduces the defining representation* of Example 3.4. Higher-degree polynomials furnish a tower of further representations — the quadratics x_1^2, x_1x_2, x_2^2 carry a three-dimensional one — whose decomposition into irreducibles is a short computation once we have the tools of Lecture 5.

Reading

The dual representation

For a general (not necessarily orthogonal) representation, the computation above does not return the same matrices, and what it does return deserves its own name. Recall that the *dual space* V^* of V is the space of linear maps $\varphi: V \rightarrow \mathbb{C}$ (*functionals*); given a basis (e_i) of V , the dual basis (e^i) of V^* is defined by $e^i(e_j) = \delta_{ij}$. Linear functionals are functions on V , so the function-space construction applies to them, and it preserves linearity:

Proposition 3.10 (Dual representation). *Let (ρ, V) be a representation. Then*

$$(\rho^*(g)\varphi)(v) := \varphi(\rho(g^{-1})v), \quad \varphi \in V^*, v \in V,$$

defines a representation (ρ^, V^*) , the dual (or contragredient) representation. In the dual basis its matrices are*

$$D^*(g) = D(g^{-1})^T = (D(g)^{-1})^T.$$

Proof. Each $\rho^*(g)$ maps functionals to functionals linearly, and the homomorphism property is Theorem 3.8 applied to the action of G on the set V via ρ . For the matrices: $(\rho^*(g)e^j)(e_i) = e^j(\rho(g^{-1})e_i) = D_{ji}(g^{-1})$, so expanding $\rho^*(g)e^j$ in the dual basis gives coefficients $D_{ij}^*(g) = D_{ji}(g^{-1})$, which is the stated matrix. One can also verify directly that $g \mapsto D(g^{-1})^T$ is a homomorphism: transposition and inversion each reverse the order of products, and two reversals cancel. ■

If the matrices $D(g)$ are *unitary* — and for a *finite group* the next section shows this can always be arranged — then $D(g^{-1}) = D(g)^\dagger$ and $D^*(g) = \overline{D(g)}$: *the dual representation is the complex conjugate representation*. The physics reader has met this contragredient behaviour before: if kets transform as $|\psi\rangle \mapsto \rho(g)|\psi\rangle$, then bras transform with the conjugate matrices so that $\langle\varphi|\psi\rangle$ stays invariant. The distinction between a representation and its conjugate, invisible when the matrices are real, prepares an important distinction in later SU(3) study: the quark triplet $\mathbf{3}$ and the antiquark antitriplet $\bar{\mathbf{3}}$ are conjugate representations. This course builds the dual/conjugate vocabulary but does not teach that particle-physics application; Jones, Ch. 8, §§8.3–8.4 is a further-reading destination.

The function-space construction is fully general, and we shall lean on it repeatedly. It is how a finite group acts on polynomials (the sign of a permutation, Lecture 1, and Example 3.9); how a point group acts on the displacement patterns of a vibrating molecule

(Lecture 6); how a symmetry group acts on quantum wavefunctions, $\psi(\vec{r}) \mapsto \psi(g^{-1}\vec{r})$, which is the formula underlying degeneracies and selection rules (Lecture 7); and how the rotation group acts on functions on the sphere, whose stable subspaces are spanned by the spherical harmonics (Part II). The same formula serves in every one of these places.

3.3 Equivalence of representations

When are two representations “the same”? As with groups themselves (Lecture 1), the answer is: when a structure-preserving dictionary translates one into the other.

Definition 3.11 (Equivalence). Representations (ρ_V, V) and (ρ_W, W) of the same group G are *equivalent*, written $\rho_V \cong \rho_W$, if there is an invertible linear map $T: V \rightarrow W$ with

$$T \rho_V(g) = \rho_W(g) T \quad \text{for all } g \in G.$$

Such a T is called an *intertwiner* (it lets the group action pass through it).

In matrix language: choosing bases and letting S be the matrix of T , equivalence reads $D_W(g) = S D_V(g) S^{-1}$ for all g , with one and the same S . In particular the matrices of a single representation in two different bases are related this way — equivalent representations are one representation written in different coordinates, and all notions we define (irreducibility, characters, multiplicities) will be invariant under equivalence.

Example 3.12 (A representation that splits over $\mathbb{C}$ but not over $\mathbb{R}$). Let $C_3 = \langle r \rangle$ act on $\mathbb{C}^2$ through the rotation matrices of Example 3.4: $D(r^k) = R(2\pi k/3)$. The vectors $v_{\pm} = \begin{pmatrix} 1 \\ \mp i \end{pmatrix}$ are eigenvectors of every rotation:

$$R(\theta) \begin{pmatrix} 1 \\ \mp i \end{pmatrix} = \begin{pmatrix} \cos \theta \pm i \sin \theta \\ \sin \theta \mp i \cos \theta \end{pmatrix} = e^{\pm i\theta} \begin{pmatrix} 1 \\ \mp i \end{pmatrix},$$

as multiplying out confirms. So in the basis (v_+, v_-) the matrices become

$$S^{-1} D(r^k) S = \begin{pmatrix} \omega^k & 0 \\ 0 & \omega^{-k} \end{pmatrix}, \quad \omega = e^{2\pi i/3},$$

where $S = (v_+ | v_-)$ is the matrix whose columns are the vectors v_+ and v_- . So the rotation representation of C_3 is equivalent, over $\mathbb{C}$, to the direct sum $\chi_1 \oplus \chi_2$ (indices mod 3, so $\chi_2 = \chi_{-1}$) of two of the one-dimensional representations of Example 3.3 — the *direct sum* $\rho \oplus \sigma$ of two representations being the block-diagonal action $g \mapsto \rho(g) \oplus \sigma(g)$ on $V \oplus W$. Notice that the diagonalising basis is genuinely complex: over the *reals* a rotation through $2\pi/3$ has no eigenvector at all, and the representation cannot be split. Working over $\mathbb{C}$ is not a convenience but a structural choice, and it is the right one for quantum mechanics, where state spaces are complex from the outset.

3.4 Invariant subspaces and irreducibility

Example 3.12 displayed a representation containing smaller ones. The general notions:

Definition 3.13 (Invariant subspace; irreducibility). Let (ρ, V) be a representation. A subspace $W \subseteq V$ is *invariant* if

$$\rho(g)W \subseteq W \quad \text{for all } g \in G.$$

Equivalently, $\rho(g)W = W$ (it suffices to act on the condition above with $\rho(g^{-1})$). An invariant subspace carries the *subrepresentation*

$$\rho|_W : g \mapsto \rho(g)|_W.$$

The representation is *irreducible* if $\dim V \geq 1$ and its only invariant subspaces are $\{0\}$ and V ; otherwise it is *reducible*. Every one-dimensional representation is irreducible.

Example 3.14 (Inside the permutation representation). In the permutation representation of S_n on $\mathbb{C}^n$ (Example 3.6), the line

$$L = \mathbb{C}u_0, \quad u_0 = e_1 + e_2 + \cdots + e_n,$$

is invariant: permuting the summands leaves u_0 fixed, so L carries a copy of the trivial representation. A second invariant subspace is

$$V_0 = \left\{ v \in \mathbb{C}^n : \sum_i v_i = 0 \right\},$$

of dimension $n - 1$: permuting the components of v does not change their sum. So the permutation representation is reducible for every $n \geq 2$, and $\mathbb{C}^n = L \oplus V_0$ as vector spaces, both summands invariant. The subrepresentation on V_0 is called the *standard representation* of S_n ; for $n = 3$ the tutorial proves it irreducible.

In the example, the invariant subspace came with an invariant complement. That this is automatic for finite groups is the content of Maschke's theorem — and it is genuinely a theorem, not an algebraic triviality:

Example 3.15 (Reducible but indecomposable). Let the *infinite* group $(\mathbb{Z}, +)$ be represented on $\mathbb{C}^2$ by the operators whose matrices in the basis (e_1, e_2) are

$$D(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \quad D(n)D(m) = \begin{pmatrix} 1 & n+m \\ 0 & 1 \end{pmatrix} = D(n+m).$$

The line $\mathbb{C}e_1$ is invariant, so the representation is reducible. But no *complementary* invariant line exists: any line other than $\mathbb{C}e_1$ is spanned by some $\begin{pmatrix} a \\ 1 \end{pmatrix}$, and

$$D(1)\begin{pmatrix} a \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a \\ 1 \end{pmatrix} = \begin{pmatrix} a+1 \\ 1 \end{pmatrix}$$

is never proportional to $\begin{pmatrix} a \\ 1 \end{pmatrix}$. The representation cannot be written as a direct sum of smaller ones — the matrices can be made simultaneously triangular but never simultaneously block-diagonal. Finiteness of the group is exactly what will rule this pathology out.

3.5 Unitarity: the averaging trick

Equip V with a Hermitian inner product $\langle \cdot, \cdot \rangle$, conjugate-linear in the first (left) argument (the physics convention). An operator U is *unitary* if $\langle Uu, Uv \rangle = \langle u, v \rangle$ for all $u, v \in V$, and a representation is *unitary* (with respect to the given product) if every $\rho(g)$ is.

Theorem 3.16 (Unitarisability). *Let (ρ, V) be a representation of a finite group G , and let $\langle \cdot, \cdot \rangle_0$ be any inner product on V . Then*

$$\langle u, v \rangle := \frac{1}{|G|} \sum_{h \in G} \langle \rho(h)u, \rho(h)v \rangle_0$$

is an inner product on V with respect to which ρ is unitary.

Proof. Sesquilinearity (conjugate-linearity in the first argument and linearity in the second) is inherited term by term, and positivity holds because a sum of nonnegative terms vanishes only if each term does — in particular the $h = e$ term $\langle u, u \rangle_0$, so $\langle u, u \rangle = 0$ forces $u = 0$.

For unitarity, fix g and compute

$$\langle \rho(g)u, \rho(g)v \rangle = \frac{1}{|G|} \sum_{h \in G} \langle \rho(hg)u, \rho(hg)v \rangle_0 = \frac{1}{|G|} \sum_{h' \in G} \langle \rho(h')u, \rho(h')v \rangle_0 = \langle u, v \rangle,$$

because as h runs once over G , so does $h' = hg$: the map $h \mapsto hg$ is a bijection of G (cancellation — the Latin-square property of the Cayley table, Lecture 1). ■

Remark 3.17 (The averaging trick). The step the proof turns on — *average a quantity over the group, and the average is automatically invariant because translating the summation variable merely reshuffles the sum* — is the most frequently reused idea in this subject. It will prove Maschke's theorem in a moment, the orthogonality relations in Lecture 4, and the projection formulae in Lecture 5; and in Part II, with the sum replaced by an invariant integral over a compact Lie group, the entire theory will be rebuilt on it.

Theorem 3.16 unitarises the *product*: it leaves the matrices alone and changes what “unitary” means. In practice one wants the other direction — unitary *matrices*, in a fixed and ordinary inner product — and the passage between the two is explicit.

Fix a basis and let M be the Gram matrix of the invariant product, $M_{ij} = \langle e_i, e_j \rangle$, so that

$$\langle u, v \rangle = \sum_{i,j=1}^n \bar{u}_i v_j \langle e_i, e_j \rangle = \sum_{i,j=1}^n \bar{u}_i M_{ij} v_j = u^\dagger M v.$$

Then M is Hermitian and positive definite, and invariance of the product says exactly

$$D(g)^\dagger M D(g) = M \quad \text{for all } g \in G.$$

Corollary 3.18 (Unitarising the matrices). *Let D be a matrix representation of a finite group G and let M be as above. Let S be any invertible matrix with*

$$S^\dagger M S = \mathbb{1},$$

that is, any matrix whose columns are a basis orthonormal for the invariant product. Then

$$U(g) := S^{-1} D(g) S$$

is unitary for every $g \in G$, and U is a matrix representation equivalent to D . So every matrix representation of a finite group is equivalent to a unitary one.

Proof. Rearranging $S^\dagger M S = \mathbb{1}$ gives $M = (S^{-1})^\dagger S^{-1}$, so

$$U(g)^\dagger U(g) = S^\dagger D(g)^\dagger (S^{-1})^\dagger S^{-1} D(g) S = S^\dagger (D(g)^\dagger M D(g)) S = S^\dagger M S = \mathbb{1}.$$

That U is a representation, and equivalent to D , is Definition 3.11 with intertwiner S^{-1} . ■

How to find S . Apply Gram–Schmidt to the standard basis $(e_1, \dots, e_n)$ in the invariant product and take the results as the columns of S . Because Gram–Schmidt builds the j -th vector out of the first j , this S is upper triangular, and $M = (S^{-1})^\dagger S^{-1}$ is the Cholesky factorisation of M . It is convenient to compute with a computer algebra system.

Tutorial

Example 3.19 (Averaging, concretely). Let $C_2 = \{e, a\}$ act on $\mathbb{C}^2$ by

$$D(e) = \mathbb{1}, \quad D(a) = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \quad D(a)^2 = \mathbb{1} \checkmark,$$

a representation that is *not* unitary for the standard product: $D(a)^\dagger D(a) = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \neq \mathbb{1}$. Average the standard product $\langle u, v \rangle_0 = u^\dagger v$ over both group elements:

$$\langle u, v \rangle = \frac{1}{2} \left(\langle D(e)u, D(e)v \rangle_0 + \langle D(a)u, D(a)v \rangle_0 \right) = \frac{1}{2} \left(u^\dagger v + u^\dagger D(a)^\dagger D(a) v \right) = u^\dagger M v,$$

where

$$M = \frac{1}{2} \left(\mathbb{1} + D(a)^\dagger D(a) \right) = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 3/2 \end{pmatrix}.$$

One verifies $D(a)^\dagger M D(a) = M$: the representation is unitary for the new (invariant) product.

The same thing is visible on passing to the eigenbasis of $D(a)$. The matrix $D(a)$ has two eigenvalues, $+1$ and -1 , with eigenvectors $v_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $v_- = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$. The vectors $v_\pm$ are orthogonal in the *invariant* product though not in the standard one:

$$v_+^\dagger M v_- = (1 \ 0) \begin{pmatrix} 1 & 1/2 \\ 1/2 & 3/2 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = (1 \ 0) \begin{pmatrix} 0 \\ -5/2 \end{pmatrix} = 0.$$

In the basis (v_+, v_-) the matrix $D(a)$ becomes $\text{diag}(1, -1)$ — manifestly unitary, and incidentally displaying the decomposition of this representation into the two one-dimensional representations of C_2 .

That route used eigenvectors, and it worked only because this representation happens to split into one-dimensional pieces. The recipe of Corollary 3.18 needs none: Gram–Schmidt (e_1, e_2) in the invariant product. Since $\langle e_1, e_1 \rangle = 1$ the first vector is already $f_1 = e_1$, and with $\langle f_1, e_2 \rangle = \frac{1}{2}$,

$$f'_2 = e_2 - \frac{1}{2}e_1 = \begin{pmatrix} -1/2 \\ 1 \end{pmatrix}, \quad \langle f'_2, f'_2 \rangle = \frac{5}{4}, \quad f_2 = \frac{2}{\sqrt{5}} \begin{pmatrix} -1/2 \\ 1 \end{pmatrix}.$$

Taking these as columns,

$$S = (f_1 \mid f_2) = \begin{pmatrix} 1 & -1/\sqrt{5} \\ 0 & 2/\sqrt{5} \end{pmatrix}, \quad S^{-1} D(a) S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the same unitary matrix as before: f_2 is v_- normalised in the invariant product, up to sign. The inverse of S

$$S^{-1} = \begin{pmatrix} 1 & 1/2 \\ 0 & \sqrt{5}/2 \end{pmatrix}$$

is upper triangular and satisfies $M = (S^{-1})^\dagger S^{-1}$ — the Cholesky factor, as the corollary says it must be.

The structural consequence of unitarity is this:

Corollary 3.20. *In a unitary representation, the orthogonal complement of an invariant subspace is invariant.*

Proof. Let W be invariant and $w' \in W^\perp$. For any $g \in G$ and $w \in W$,

$$\langle \rho(g)w', w \rangle = \langle w', \rho(g)^{-1}w \rangle = \langle w', \rho(g^{-1})w \rangle = 0,$$

the first step by unitarity and the last because $\rho(g^{-1})w \in W$. Hence $\rho(g)w' \in W^\perp$. ■

3.6 Maschke's theorem: complete reducibility

Theorem 3.21 (Maschke). *Every finite-dimensional complex representation of a finite group is a direct sum of irreducible subrepresentations: there are invariant subspaces $W_1, \dots, W_k$, each irreducible, with*

$$V = W_1 \oplus W_2 \oplus \dots \oplus W_k.$$

Proof. Induct on $\dim V$. If $\dim V = 0$ the statement is empty (the direct sum of no representations), so assume $\dim V \geq 1$. If V is irreducible (in particular if $\dim V = 1$), there is nothing to prove.

Otherwise, equip V with an invariant inner product (Theorem 3.16) and let W be an invariant subspace with $0 \neq W \neq V$. By Corollary 3.20, $V = W \oplus W^\perp$ with both summands invariant and of strictly smaller dimension; apply the induction hypothesis to each. ■

In matrix language: *there is a single invertible S such that*

$$S D(g) S^{-1} = \begin{pmatrix} D^{(i_1)}(g) & & \\ & D^{(i_2)}(g) & \\ & & \ddots \end{pmatrix} \quad \text{for every } g \in G$$

simultaneously, with each diagonal block an irreducible matrix representation (Figure 3.1). The force of the statement lies in the word *simultaneously*: there is one basis in which every group element takes the same block form at the same time. Finding that basis for representations of physical interest is, in a practical sense, what Lectures 5 and 6 are about.

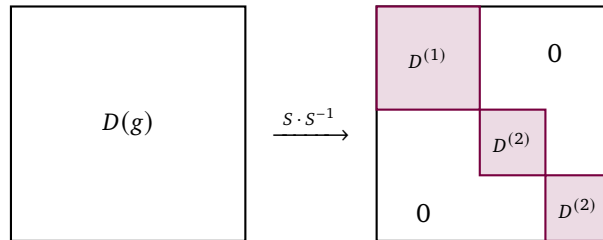


Figure 3.1: Maschke's theorem in matrix form: a single change of basis S brings every $D(g)$, for all $g \in G$ at once, to the same block-diagonal form with irreducible blocks. Blocks may repeat: here the irreducible representation $D^{(2)}$ occurs with multiplicity two.

Remark 3.22 (Notation, and what remains to prove). Collecting equivalent blocks, we write

$$V \cong \bigoplus_i m_i V^{(i)}, \quad \rho \cong \bigoplus_i m_i \rho^{(i)},$$

where $V^{(i)}$ runs over the inequivalent irreducible representations of G and the integer $m_i \geq 0$ is the *multiplicity* of $V^{(i)}$ in V . We reserve $n_i = \dim V^{(i)}$ for the irreducible dimension, so the total-dimension check is $\dim V = \sum_i m_i n_i$.

Two caveats keep us honest. First, the decomposition into subspaces is *not unique* (already two copies of the trivial representation can be resliced into different pairs of invariant lines); what will turn out to be unique is the multiset of blocks — the multiplicities m_i . Second, nothing yet tells us how many irreducibles a given group has, or how to find the m_i without hunting for invariant subspaces by hand. Lectures 4 and 5 answer all of this precisely: G has finitely many irreducibles, as many as conjugacy classes, of dimensions satisfying $\sum_i (\dim V^{(i)})^2 = |G|$, and the multiplicities are computed by a one-line character formula.

Remark 3.23 (What each hypothesis is for). Both hypotheses of Maschke's theorem are necessary. Drop finiteness of the group and Example 3.15 breaks the conclusion. Drop the field assumption and it breaks too: over fields of prime characteristic p dividing $|G|$, the averaging factor $1/|G|$ does not exist and complete reducibility fails; what survives in that case is the subject called modular representation theory. Over $\mathbb{R}$, by contrast, the proof goes through verbatim with real inner products — though, as Example 3.12 showed, the *irreducible pieces* over $\mathbb{R}$ and over $\mathbb{C}$ can differ. In Part II, compactness of the Lie group will play the role finiteness plays here.

Tutorial

3.7 Worked example: the permutation representation of S_3

We now execute the whole programme by hand on the three-dimensional permutation representation of S_3 (Example 3.6), with its invariant subspaces $L = \mathbb{C}u_0$ and $V_0 = \{\sum_i v_i = 0\}$ from Example 3.14. Since $(1\ 2)$ and $(1\ 2\ 3)$ generate S_3 , every identity we need can be checked on these two elements alone.

Step 1: adapt the basis. Choose

$$u_0 = e_1 + e_2 + e_3, \quad f_1 = e_1 - e_2, \quad f_2 = e_2 - e_3,$$

so that (u_0, f_1, f_2) is a basis of $\mathbb{C}^3$ with u_0 spanning L and (f_1, f_2) spanning V_0 .

Step 2: compute the blocks. The transposition $(1\ 2)$ swaps e_1, e_2 and fixes e_3 , so

$$f_1 \mapsto e_2 - e_1 = -f_1, \quad f_2 \mapsto e_1 - e_3 = f_1 + f_2;$$

the cycle $(1\ 2\ 3)$ sends $e_1 \mapsto e_2 \mapsto e_3 \mapsto e_1$, so

$$f_1 \mapsto e_2 - e_3 = f_2, \quad f_2 \mapsto e_3 - e_1 = -f_1 - f_2.$$

Reading off columns, the matrices of the subrepresentation on V_0 in the basis (f_1, f_2) are

$$B((1\ 2)) = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B((1\ 2\ 3)) = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix},$$

and indeed $B((1\ 2))^2 = 1$, $B((1\ 2\ 3))^3 = 1$. In the full basis (u_0, f_1, f_2) , with

$$S = (u_0 \mid f_1 \mid f_2) = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix}$$

the change-of-basis matrix, we obtain in block form

$$S^{-1}D(g)S = \begin{pmatrix} 1 & 0 \\ 0 & B(g) \end{pmatrix} \quad \text{for all } g \in S_3$$

— block-diagonal form achieved, with blocks of sizes 1 and 2, as direct multiplication confirms.

Step 3: the 2×2 block is irreducible. A proper nonzero invariant subspace of V_0 would be a line, spanned by a common eigenvector of all the $B(g)$ — in particular of both generators. The eigenvectors of $B((1\ 2))$ (eigenvalues -1 and $+1$) are the multiples of

$$w_- = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad w_+ = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

and neither line survives the cycle:

$$B((1\ 2\ 3))w_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \nparallel w_-, \quad B((1\ 2\ 3))w_+ = \begin{pmatrix} -2 \\ -1 \end{pmatrix} \nparallel w_+.$$

No common eigenvector, no invariant line: the standard representation of S_3 is irreducible. Altogether

$$\mathbb{C}^3 \cong V^{\text{triv}} \oplus V^{\text{std}},$$

the trivial representation plus a two-dimensional irreducible one, each with multiplicity one — and the sign representation, a third irreducible, does not appear at all. That these three exhaust the irreducibles of S_3 is proved in Lecture 4; here we know only that they are three inequivalent ones. Lecture 5 will read the multiplicities 1, 1, 0 off in one line, with no hunting for invariant subspaces at all.

Remark 3.24 (The geometry of the standard representation). The decomposition has a picture. S_3 permutes the coordinate axes of $\mathbb{R}^3$; the diagonal direction u_0 is fixed, and the action turns the perpendicular plane V_0 into itself. The orthogonal projections of e_1, e_2, e_3 onto that plane form an equilateral triangle, which the group permutes — so the standard representation is nothing but the defining representation of $D_3 \cong S_3$ (Lecture 1) in disguise: the abstract block B and the geometric matrices of Example 3.4 are equivalent. One group, the same irreducible two-dimensional representation, two coordinate descriptions.

Summary.

- *Representation.* A homomorphism $G \rightarrow \text{GL}(V)$, equivalently a linear action of G on V (Definition 3.1).
- *Function space.* G acts on functions by $(\rho(g)\psi)(x) = \psi(g^{-1}x)$ (Theorem 3.8); the reading carries the dual representation on the dual space.
- *Irreducibility.* An invariant subspace is one preserved by G ; a representation is irreducible when it has no proper invariant subspace (Definition 3.13).

- *Unitarity*. Every representation of a finite group is unitarisable by averaging an inner product (Theorem 3.16).
- *Maschke*. Every finite-dimensional representation of a finite group is a direct sum of irreducibles (Theorem 3.21).

Tutorial. Exercise Sheet 3.

References

Jones, *Groups, Representations and Physics*, Ch. 3; Tung, *Group Theory in Physics*, §§3.1–3.3; Fulton–Harris, *Representation Theory*, §§1.1–1.2; Serre, *Linear Representations of Finite Groups*, §§1.1–1.4.